

The “Bosch-A radar” object bank is the camera’s output

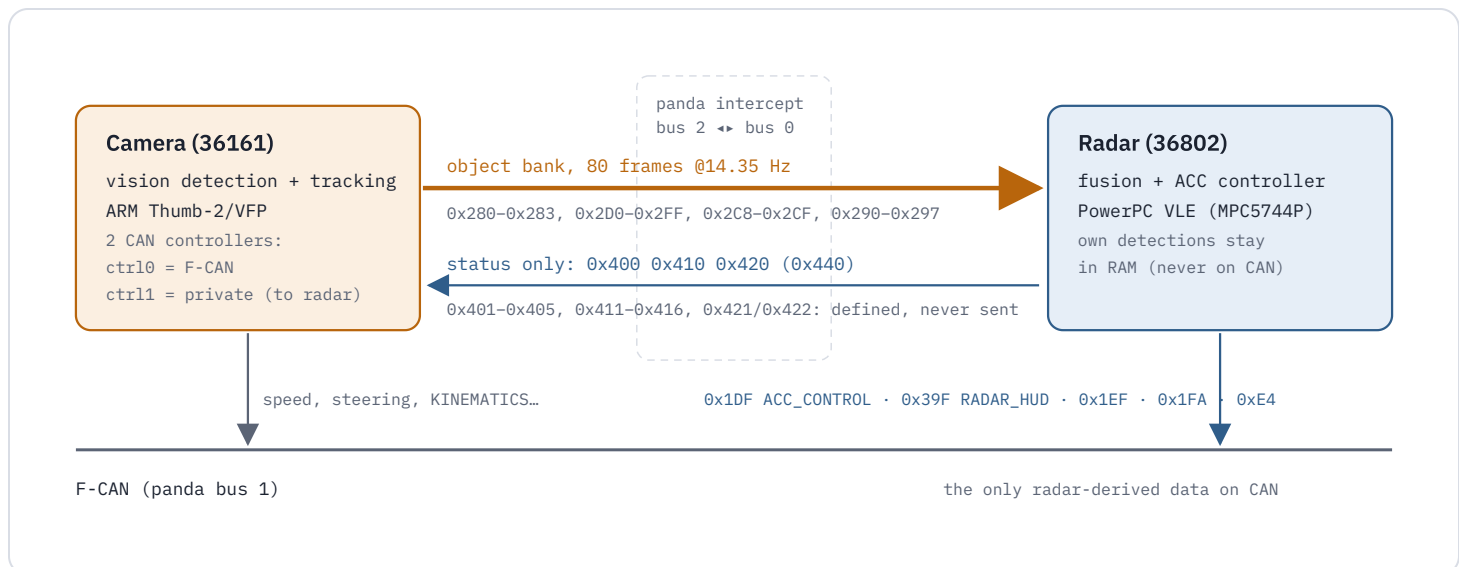
On Honda Bosch cars, the 16-slot object list openpilot reads as radar points (`0x280–0x2FF` , `0x2C8–0x2CF` , `0x290–0x297`) is produced and transmitted by the windshield camera. The radar receives that list and fuses it with its own measurements internally. Its own detections never appear on CAN.

Scope: Civic 10th gen (camera 36161-TBA-A130, radar 36802-TBA-A150), CR-V 5G (TLA), CR-V Hybrid (TPG). Firmware: camera 36161-TLA-A070 image (decoded), radar 36802-TBA image.

Bottom line. Every “radar” characterization we have done was a characterization of Honda’s camera tracker. That includes U11 lag and shrink, range quality, the scale fits, and the existence and sigma fields. Every “radar vs vision” cross-check was really vision vs vision.

The 1/72 velocity decode and the 1/16 m range decode are still correct: they are the camera’s own encoding.

How the data actually flows



The panda sits on the private camera↔radar CAN. Bus 2 is the camera side, bus 0 the radar side. Bus 1 is F-CAN.

Evidence

Five independent lines point the same way. Each one is labelled with how strong it is.

LINE	WHAT WAS FOUND	STRENGTH
Camera firmware: TX	The camera's transmit PDU table lists all 80 bank IDs, alongside 0xE4 steering, LKAS HUD and its other outputs. The generated packers write every bank field (range, azimuth, U11, U10, existence, lifecycle...). All 16 slots map to exactly the IDs openpilot parses.	PROVEN
Camera firmware: RX	The CAN receive handler and its tables decode to exactly two controllers. Controller 0 is F-CAN. Controller 1 is the private radar bus, carrying 0x256 plus the radar family 0x400–0x405, 0x410–0x416, 0x420–0x422 and 0x440. There is no third bus.	PROVEN
Radar firmware	The radar's receive filters include the 0x280 bank. Its 25 transmit PDUs each pack single values from one internal context object; none is a track list. The three it sends to the camera carry status bits, status bytes and an ASCII ID. Two unsent groups line up with the camera's unused RX slots by length: 0x401–0x405 (7,5,7,2,5) and 0x411–0x416 (7,6,6,7,6,6).	PROVEN
Wire, Civic	Across 416 segments, stock and openpilot longitudinal, the radar side of the private bus carries only 0x400, 0x410 and 0x420, with 4–8 distinct payloads each. Bus 2 carries the camera's full TX table.	OBSERVED
Wire, CR-V 5G and CR-V Hybrid	Public openpilot CI logs, stock ACC: the radar sends only 0x400, 0x410 ("TLAA1..." / "TPGA0..."), 0x420 and 0x440. The camera sends the 80-frame bank with sentinels identical to the Civic's.	OBSERVED
Field semantics	The fields are vision quantities: left/right edges and angular width (a bounding box), <code>NORMALIZED_CLOSING</code> = $1/\text{TTC}$, <code>RANGE_RATIO</code> = $\text{range}(t-1)/\text{range}(t)$ (scale change), detection existence, and a field of view of about ± 0.39 rad.	CONSISTENT
Behaviour	At matched speeds the bank reports $1.5\text{--}3.3\times$ fewer stationary objects at night, while tracking more moving cars, farther out (see "Low-light test"). U11 behaves like openpilot's own vision velocity (ratio ≈ 1.1) and lags about 0.56 s on every object.	CONSISTENT

Low-light test

A radar measures with its own transmitted signal, so ambient light should make no difference to what it detects. A camera depends on light. Unlit stationary objects, such as parked cars without lights, poles, barriers and signs, should become hard to see at night. Moving cars stay easy because their lights are on. So the question was: does the object bank behave differently in the dark, and in which way?

Method

- **Data:** Peter’s Civic, 416 one-minute segments over 13 routes. Raw camera-side CAN was decoded independently of the openpilot parser.
- **Day or night per segment:** sun elevation computed from the GPS time and position (north Texas, about 33°N). *Day* means sun above 15°, *night* means sun below −6° (after civil twilight). The road camera’s exposure value confirms the split: median 0.0 % by day and 9.5 % at night.
- **Active object:** STATUS ≠ 15 with a valid range.
- **Stationary-like object:** the camera’s own velocity U11 within ±10 % of −(ego speed), with ego speed at least 8 m/s, U11 not on its rail and U10 < 128. Above about 12 m/s a stationary object rails U11 and cannot be counted, so the test effectively uses moments with ego speed 8–12 m/s.
- **Segments used:** those with enough data (over 500 active object sweeps, or over 200 with the ego moving, for the totals).

Result 1: what the bank sees, per segment (medians)

CONDITION	SEGMENTS	CAMERA EXPOSURE	OBJECTS PER SWEEP	RANGE P95	SHARE OF OBJECTS BEYOND 80 M	STATIONARY-LIKE SHARE
Day (sun > 15°)	182	0.0 %	1.96	105.7 m	15.9 %	2.1 %
Night (sun < −6°)	121	9.5 %	2.46	112.1 m	23.5 %	0.0 %

At night the bank tracks *more* objects and sees *farther*, but the median night segment contains no stationary-like objects at all. Lit, moving vehicles are easier for a camera at night (headlights and taillights stand out); unlit stationary things are not.

Result 2: stationary-like objects in total, with speed matching

CONDITION	SEGMENTS	OBJECT SWEEPS (EGO MOVING)	STATIONARY-LIKE	RATE	SEGMENTS WITH ANY	MEAN EGO SPEED
Day	152	242,193	6,168	2.55 %	91 of 152 (60 %)	14.1 m/s
Night	113	234,893	1,365	0.58 %	24 of 113 (21 %)	21.6 m/s
Day, segment mean speed 8–12 m/s	33			6.36 %		
Night, segment mean speed 8–12 m/s	11			4.37 %		
Day, segment mean speed 12–30 m/s	96			1.22 %		
Night, segment mean speed 12–30 m/s	84			0.37 %		

Overall the stationary-like rate is 4.4× lower at night. Night driving here is faster on average, and faster driving means fewer qualifying moments, so the speed-matched rows are the fairer comparison. They still show fewer stationary-like objects at night:

- **1.5× fewer** on slower segments (8–12 m/s mean speed, mostly town driving, where stopped cars with lit taillights are common)
- **3.3× fewer** on faster segments (12–30 m/s, where stationary things are mostly roadside objects with no lights)

How to read it

- **The pattern fits a camera and not a radar:** unchanged or better detection of lit moving vehicles, with a sharp drop in stationary objects. A radar would show no day/night difference for either.

- **The surviving night stationary objects** are concentrated in slow, town-like driving, where they are most plausibly stopped cars with taillights on or objects lit by streetlights.

Caveats

- **Road mix is not controlled:** night drives may use different roads (more highway, fewer parked cars). Speed matching reduces this but does not remove it.
- **The classifier uses the camera’s own velocity output**, so it relies on U11 being right for stationary objects. That was checked separately: 0.98 against GPS and 1.00 against odometry.
- **A segment is one minute**, labelled from its first valid GPS fix. Dusk and dawn segments (sun between -6° and 15°) are excluded.
- **This is behavioural support, not proof.** The proof is the firmware and wire evidence above; this test only shows that the bank behaves the way a camera would.
- **A stronger version:** the same route driven by day and by night, with the stationary objects identified (parked car vs roadside structure) from the video.

What the camera firmware says about each field

The camera image was decrypted with `plain = (0x17 - ((0x03 - enc) ^ 0x10)) & 0xFF`. That is the RWD header keys with *reversed* subtraction, which rwd-xray’s key search cannot express. The embedded 16-entry part-number list confirms it. Every packer bit position matches the DBC.

FIELD	ENCODING IN THE CAMERA	WHAT IT MEANS FOR THE PARSER
U11 (AUX 7 11)	<code>rne((clamp(v,-12,12) + 12) / f32(1/72)) ;</code> 0x7FE = default/sentinel	/72 decode is exact; rails ± 12.0 m/s
Range (F0 23 12)	<code>r / 0.0625</code> , clamp 0–255.875, 0xFFF sentinel, no offset	1/16 m is exact. The radar’s $-n/128$ belongs to its own lateral computation ($AC004 = \tan f(az) \cdot (q16 - n)/128$), not to the wire format
Existence (F1 46 7)	<code>(p - 1/64) · 128</code> , $p \in [1/64, 1]$	Valid 0–126; 127 = not available . It is never seen on an active object (0 of 626,100)

FIELD	ENCODING IN THE CAMERA	WHAT IT MEANS FOR THE PARSER
Lifecycle (F2 7 12)	clamped to $\leq 0xFFE$; $0xFFFF$ reserved	Saturation at $0xFFE$ is real (confirms the PR #17 fix)
Range sigma (F0 7 7)	$\sigma / 0.25$ m, sentinel 127	The parser's " ≥ 4 " threshold means $\sigma \geq 1.0$ m
Angular width (F1 23 11)	$(\text{edgeA} - \text{edgeB}) \cdot 2048$ rad	The DBC's 0.03125 deg scale is wrong; $1/2048$ rad = 0.02798 deg
U10 (AUX 23 10)	$\cdot 144$, clamp 0–865 in the TLA image	The TBA build exceeds 865, so Peter's camera is a different software revision of the same family

What this explains

U11 lag and “shrink”

U11 is a tracker velocity, a low-pass with $\tau \approx 0.6$ s. Against the bank's own range it reads about 0.72× on the in-lane lead. Stopped cars and stationary objects are read correctly: 0.98 against GPS and 1.00 against ego odometry.

“Radar beats vision”

This really means Honda's vision tracker is smoother than openpilot's model. It also means the 0.98 agreement between “radar” range rate and openpilot's vision distance is not ground truth.

Range errors near cars

Behind stopped cars, bank range closes only 71–90 % of ego odometry. Against openpilot's model it runs +2.3 m at 5–10 m and –2.7 m at 70–90 m. That pattern is typical of monocular calibration. Which system is off is unresolved.

Lead vs other cars

Adjacent-lane cars have U11 and range in agreement (1.01 closing). The in-lane lead does not (0.90 closing, 0.53 opening). The camera handles its in-path object differently; it is not radar input.

What it does *not* change

- **The decodes stand.** U11 at 1/72, range at 1/16 m, azimuth at 1/2048 rad and the sentinels are firmware-exact.
- **The bank is still useful.** It is a second, independent vision tracker that is smooth and long-range for vehicles. It just isn't radar, and its errors can be correlated with openpilot's own vision in the same scenes.
- **The rail-is-a-bound and never-delete-a-point safety rules still apply.** They are about not discarding evidence, whatever the sensor.

Implications

- **On these cars openpilot has no radar data at all.** Treat radar's "radar leads" as camera tracks when deciding how much to trust them against the model.
- **Range-vs-U11 corrections are camera-vs-camera corrections.** These include D-053, PR #17's range assist, and the newer range-correction filter. Validate them against an independent reference: odometry against stopped targets, or tape/laser ground truth. Do not validate them against openpilot vision.
- **Vision-speed bounds lean on the weakest channel.** This covers far-rail, vision-assist and camera x-rate. The model's lead *speed* under-reads closing even more than U11; its *distance* rate is the better vision signal.
- **Parser wording should change.** "The radar rates it as barely there" and similar comments describe the camera. Existence 127 should be published as "not provided".

Still open

QUESTION	HOW TO SETTLE IT	STATUS
Is the camera's range calibrated correctly on this car?	Tape or laser to a parked car at 5, 10, 20, 40 and 60 m, compared with logged range; Honda camera-aiming status	OPEN
Can we get the radar's fused lead?	Decode RADAR_HUD <code>LEAD_DISTANCE</code> and ACC_CONTROL on stock-ACC drives (the raw values seen so far, 110–157, are not plain meters)	OPEN

QUESTION	HOW TO SETTLE IT	STATUS
Can we get real radar detections?	Read the radar's RAM object pools over a diagnostic or calibration memory read; the pools are already mapped in the radar firmware	PROJECT
What do 0x401–0x405 carry when enabled?	Trace the radar's PDUs 14–18 fill functions (signals 0xA7–0xC5); the camera already routes them to the same RX path	OPTIONAL
Exact TBA camera image	Not in any available archive; Civic findings rest on family evidence plus wire behaviour	OPEN

Reproducing this

- **Camera image:** 36161TLA_A070M1__A1710290.rwd (firmware-analysis-kit, [iHDS_rwds/CalibFiles](#)); RWD SHA-256 c445d80d...c1f; decoded payload SHA-256 d2528ea5...b155.
- **Camera addresses** (payload offsets): U11 formatter [0x143C64](#); slot builder [0x133796](#); TX PDU table [0x1AD3E4](#); CAN RX handler [0x1214F8](#); HRH table [0x1ADB74](#); RX ID table [0x1ADE48](#). Config is linked at offset – 0xC0000.
- **Radar image:** 36802TBA (A160), SHA-256 aaf96045...52bb; TX PDU table at file [0x134C98](#); fills for 0x400/0x420/0x410 at chip [0xDE272](#) / [0xDE11E](#) / [0xDE1FE](#).
- **CR-V logs** (public openpilot CI): 1ad763dd22ef1a0e/2020-02-29--18-37-03 segments 1–2 (CR-V 5G) and 14fed2e5fa0aa1a5/2021-05-25--14-59-42 segment 1 (CR-V Hybrid).
- **Full report and scripts:** [docs/research/pr17-bosch-radar-followup-2026-10-03.md](#), Addenda 6–9, and its [scripts/](#) folder.

Static firmware analysis plus logged-data analysis. Nothing was flashed or modified, and no road validation was done.